

Table 1: Response quality degradation under emotion misperception (Δ = Aligned–Conflict; higher = greater degradation). 1–5 Likert; $n=4$ per emotion, $N=20$.

Emotion	Degradation (Δ)		
	Faith.	Emp.	Rel.
Angry	+0.75	0.00	0.00
Happy	+1.75	+3.00	+2.50
Sad	0.00	0.00	0.00
Neutral	+1.50	+2.00	+0.75
Surprised	+2.50	+1.00	+2.00
Overall	+1.30	+1.20	+1.05

1 Observation

1.1 Feasibility and Asymmetry of Audio Emotion Flipping

1.2 Downstream Impact of Emotion Misperception

Beyond the logit-level vulnerability established in Section 1.1, we ask whether emotion misperception propagates into the model’s downstream response behavior. We design an attack-agnostic experiment: for each audio sample, the model (Voxtral-Mini-3B [?]) receives an *Aligned* prompt (truthful emotion description) and a *Conflict* prompt (false emotion with maximum affective contrast). We evaluate $N = 20$ samples from ESD [?] across five emotions. An independent LLM judge (DeepSeek V3 [?]) rates each response on Faithfulness, Empathy, and Relevance (1–5 Likert).

(1) Misperception causes systematic response degradation with pronounced emotion asymmetry. As shown in Table 1, the Conflict condition degrades all three dimensions (overall Δ : Faithfulness +1.30, Empathy +1.20, Relevance +1.05), but the effect is highly asymmetric across source emotions. *Happy* is the most vulnerable (Δ Empathy = +3.00, Δ Relevance = +2.50), while *sad* is completely robust ($\Delta = 0$ across all dimensions)—the model even explicitly overrides the misleading prompt in some cases, stating “*They are not angry or frustrated, but rather expressing their grief and pain.*” This valence-dependent asymmetry, where positive emotions are more susceptible than negative ones, converges with the logit-level fragility of happy identified in Section 1.1.

(2) Misperception induces hallucination in emotionally ambiguous inputs. Neutral audio exhibits a distinct failure mode: the model fabricates emotional narratives with no basis in the audio. For a factual statement (“*The package is scheduled to arrive tomorrow afternoon*”) paired with an anger prompt, the model attributes frustration to the speaker, constructs causal explanations, and generates coping strategies—none grounded in the audio content (Faithfulness drops to 1.50/5.00). This indicates that when audio carries weak emotional signals, the model defaults to the text prompt as its primary emotion source, making emotionally neutral speech—the most common category in real-world deployments—a particularly vulnerable attack surface.

1.3 Insufficiency of Traditional SER Attack Methods

We test whether existing SER adversarial attack methods transfer to ALLMs. We adapt STAA-Net [?], a representative generator-based SER attack (Wave-U-Net generator, C&W untargeted loss [?], sparsity mask), to target Voxtral-Mini-3B with full white-box access. The setup deliberately favors the traditional method: the task is *untargeted* (strictly easier than targeted flipping) and the

Table 2: STAA-Net adapted to Voxtral-Mini-3B.

Parameter	Value
Attack type / Access	Untargeted / White-box
Perturbation budget (L_∞)	0.03
Train / Test samples	200 / 200 (ESD)
Mean $\ \delta\ _\infty$	0.030 (budget saturated)
Mean SNR	13.4 dB
Predictions unchanged	195 / 200 (97.5%)

perturbation budget $\varepsilon = 0.03$ (L_∞) is substantially larger than that of our ALLM-native attacks. Table 2 summarizes the configuration and results.

Despite the generator saturating its perturbation budget and converging normally during training, 97.5% of test samples produce *identical* predictions under clean and adversarial audio. The failure is not due to insufficient perturbation but to architectural incompatibility: SER attacks target standalone classifiers with fixed output spaces, whereas ALLMs route audio through an encoder–adapter–LLM pipeline where emotion judgments emerge via autoregressive generation. The perturbation signal, optimized for a classifier’s decision boundary, dissipates through the LLM’s billion-parameter processing layers. This demonstrates that SER attack methods do not transfer to ALLMs, motivating the ALLM-native methodology in Section 1.4.

1.4 Internal Mechanism of Emotion Representation